

KEERTANA VAJJHALA

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EDUCATION

University of Illinois at Urbana-Champaign, Champaign, IL, USA

Expected May 2027

Master of Engineering, Electrical and Computer Engineering

Relevant Coursework: Computer Organization and Design (ECE411), Applied Parallel Programming (ECE408, CUDA), Logic Synthesis (ECE462), Introduction to VLSI (ECE425), Advanced Computer Architecture (ECE511)

PES University, RR Campus, Bangalore, India

Aug 2019 – Jun 2023

Bachelor of Technology, Electronics and Communication (spl. VLSI)

Relevant Coursework: Advanced Digital Design, Digital VLSI, Verification of Digital Systems, Advanced Hardware Acceleration
Honors & Awards: MRD Scholarship (40% Merit Scholarship – top of class); Distinction

WORK EXPERIENCE

Chicago UpDown Computing Inc.

Hardware Intern

July 2026 – Present

- Designed and integrated performance-counter registers into the PCIe BAR0 address space, enabling host-side access to per-lane NoC statistics.
- Developed and verified SystemVerilog RTL for the accelerator's NoC fabric, interconnect, and SIT arbiter (64-lane architecture).
- Root-caused and fixed a NoC memory deadlock via RTL changes to escape-VC handling.
- Built UVM verification environments and test plans for NoC and PCIe/AXI transaction testing.
- Ran bandwidth/saturation tests to characterize NoC throughput under load.

Samsung Semiconductors India Research, Bangalore, India

Senior Engineer

Aug 2023 – Aug 2025

- Designed and developed microarchitectures using Verilog and SystemVerilog for NVMe SSD controllers targeting datacenter, client, and automotive ASIC platforms, focusing on performance optimization and reliability enhancement.
- Implemented clock gating logic across design modules to reduce dynamic power consumption.
- Integrated AXI and AHB protocol interfaces with NVMe IP.
- Ran and analyzed failures for comprehensive regressions, Synthesis, DFT Lint, CDC, and LDRC validation.
- Implemented namespace lookup table logic to process data and interface with HDMA for host data transfer.
- Ensured SRAM integrity through ECC engine logic.
- Recognized as Employee of the Month and awarded SPOT recognition for driving latency reduction initiatives and ensuring high-quality releases.

Hardware Intern

Jan 2023 – Aug 2023

- Designed and implemented embedded memory controllers and UFS 4.0 interface logic for Samsung Galaxy S25 series using Verilog HDL and ASIC design methodologies.

SKILLS

- HDL Languages:** Verilog, SystemVerilog | **EDA Tools:** Design Compiler, Virtuoso, Innovus
- Verification & Simulation:** UVM, Functional Verification, Formal Verification, Testbench Development, TVM
- Digital Design:** ASIC Design, FPGA Implementation, RTL Design, Logic Synthesis, Timing Analysis, Place-and-Route
- Protocol/Interfaces:** PCIe, AXI, AHB, NVMe | **Programming:** Python, C, C++
- Specialized Areas:** VLSI Design, Digital Systems, DFT, CDC, LDRC; Familiar with DFT Flow (ATPG and Scan Insertion)

PROJECTS

DRAM Scheduling Policy Analysis – University of Illinois Urbana-Champaign

March 2026

- Implemented and validated four DRAM command scheduling policies (FR-FCFS, Oldest-First, Read-over-Write, Idle-Bank-First) plus XOR-based bank/bank-group interleaving in DRAMsim3; characterized bandwidth, latency, and row-buffer hit rate across FFT, LU, and RADIX traces.

Standard Cell Library – RISC V - PnR Flow – University of Illinois Urbana-Champaign

Feb 2026 – March 2026

- Designed a custom standard cell library from scratch in Cadence Virtuoso with a 1.7 μm cell height, then hand-laid-out the full RISC-V32I single-cycle datapath — ALU, register file, immediate generation, and control logic — bit-sliced for regular, area-efficient abutment.
- Reduced cell area and parasitic capacitance through Euler-path-based diffusion string ordering and drain/source region merging, and closed clean LVS/DRC signoff on the full custom layout.
- Packaged the verified standard cell library into Liberty (.lib) and LEF (.lef) formats, then ran automated place-and-route in Cadence Innovus to lay out the CPU controller and integrate it with the full-custom datapath.
- Extended the flow to auto-PnR the complete CPU using only the custom standard cells, then packaged and substituted a custom register file macro (.lib/.lef) into the full CPU place-and-route, replacing the reference IP.

RISC-V Out of Order Processor – University of Illinois Urbana-Champaign

Nov 2025 – Dec 2025

- Designed and implemented a RISC-V out-of-order processor in SystemVerilog featuring ERR-based register renaming, ROB, reservation stations, LSQ, and speculative execution.
- Developed GSHARE branch prediction and misprediction recovery mechanisms to minimize control hazard penalties and improve instruction throughput.
- Built a 2-way superscalar pipeline enabling dual-issue, out-of-order execution, and in-order commit with timing analysis and functional verification.

Interactive Educational Device for the Visually Impaired – PES University

Jan 2022 – Dec 2022

- Developed a portable refreshable Braille display using servo motor control and embedded C with ESP32 microcontroller.
- Achieved 57.5% power reduction, 20°C temperature decrease, and 71.6% cost reduction while maintaining 3.6 Hz refresh rate.
- ICONAT 2023: [Interactive Educational Device for the Visually Impaired | IEEE Conference Publication | IEEE Xplore](#)